Arunabha Majumder

Lab Engineer | Experimental Setups, Sensors, Calibration and Data Acquisition

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SUMMARY

I build experimental setups and keep them running. For the past three years at Aalborg University I have designed test rigs, fitted force, pressure and motion sensors to them, calibrated them against reference sensors and collected data from 12 human participants with EMG and motion capture. Before that I worked with pneumatic actuators in a national lab in India. I enjoy the practical side of research and helping students and colleagues get reliable data out of their experiments.

KEY RESULTS

- **<10%** force error after calibrating a sensorless force estimate against a reference force sensor
- **12** participants measured with EMG, motion capture and force data across 3 modes and 2 loads
- **0.3-0.78%** position error on a pneumatic muscle with the gain-scheduled PID controller I worked on

EXPERIENCE

PhD Researcher, Experimental Robotics Lab (project VIEXO) | 2023 - Present
Aalborg University, Department of Materials and Production, Aalborg, Denmark

- Got force estimates within 10% RMS of a reference force sensor by calibrating a 5-bar parallel robot with variable-stiffness actuators against a force sensor, and then used the setup to measure human arm impedance in the gait lab.
- Collected a full dataset from 12 participants across 3 assistance modes and 2 loads by preparing the protocol and running the sessions with EMG, Qualisys motion capture and force/torque sensors, with attention to participant safety and ethics approval.
- Built and maintained custom test rigs for a shoulder exoskeleton, fitting the motor, encoder, force sensors and DAQ, and fixing faults in wiring, mounting and logging so the rig gave repeatable results.
- Cut energy use by more than 25% (Best Paper Award, IFToMM ISRM 2026) with a spring plus 6 Nm motor actuator that I designed in SolidWorks, 3D printed and tested on those rigs.
- Taught university students and supervised Bachelor's students in the lab (2023 - 2025), from setting up equipment to helping them make sense of their measurements.

MTech Researcher, Pneumatic Artificial Muscles (Master's thesis) | 2020 - 2022
CSIR-Central Mechanical Engineering Research Institute (CMERI), National Laboratory, Durgapur, India

- Fabricated and assembled pneumatic artificial muscles to an existing lab design, and ran experiments on the lab's setup with valves, pressure sensors and DAQ.
- Brought position error down to 0.3-0.78%, better than a standard PID, by working on the outer PID loop of a neural-network gain-scheduled controller in MATLAB/Simulink. I collected the experimental data, fed it to the lab's existing neural network and used its output to set the PID gains (IEEE CONECCT 2022).

Bachelor's Capstone, Upper-Limb Rehabilitation Exoskeleton | 2018 - 2019
Siddaganga Institute of Technology, Tumakuru, India

- Designed, machined, assembled and tested it with users in the college workshop. Best Major Project Award and an INR 100,000 TATA Technologies grant.

SKILLS

- **Sensors and measurement:** Force/torque sensors (Forsentec), pressure sensors, encoders, EMG, Qualisys motion capture

- **Data acquisition and setups:** DAQ hardware, calibration against reference sensors, test rigs and fixtures, Arduino
- **Software:** MATLAB/Simulink, Python, LabVIEW (basic) for logging, control and data analysis
- **Design and workshop:** SolidWorks (CSWA) for fixtures and parts, 3D printing (Bambu Lab, Ultimaker Cura); machining and assembly (Bachelor's)
- **Drawing and documentation:** Diploma in Fine Arts; hand sketches, technical illustrations and clear figures for lab guides, reports and teaching
- **Languages:** English (C1, fluent), Bengali (native), Hindi (conversational)

EDUCATION

PhD, Mechatronics and Robotics (defence expected in 2027) | 2023 - Present
Aalborg University, Department of Materials and Production, Denmark

MTech (MSc level), Mechatronics | 2020 - 2022
AcSIR - CSIR-CMERI, Durgapur, India

BTech, Mechanical Engineering | 2015 - 2019
Siddaganga Institute of Technology (VTU), India

AWARDS AND PUBLICATIONS

- Best Research Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA)
- Majumder, A., et al. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." ISRM 2026, Springer.
- Majumder, A., et al. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." IEEE CONECCT, 2022.

REFERENCES

Available upon request.